Plain: script

Segmenting Metastatic Prostate Cancer in Whole-Body PSMA PET Scans with a Hybrid CNN-Mamba Network

14 main slides · 698 words · 5:10 spoken at 135 words a minute (including 0 s of clips without narration) · target 12:00



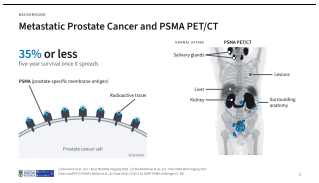
`Slide 1 · 20 s · ends at 0:20 · 23 words`

Segmenting Metastatic Prostate Cancer in Whole-Body PSMA PET Scans with a Hybrid CNN-Mamba Network

1. Purpose: Who is speaking and what the talk is about.
2. Visual: Built: title, name, supervisors and logo, with a brief entrance

Name, supervisors, UWA Medical Physics Research Group · One line: teaching a network to find prostate cancer that has spread, in whole-body PET scans

> `TRANSCRIPT TO READ · TIMED AT 30 S (TARGET 20 S)`
>
> ***"Hello everyone, thank you all for coming. My name is James Wigfield, and this project has been about training a hybrid CNN-Mamba architecture to segment metastatic prostate cancer in whole-body PSMA PET scans. I'd like to thank my supervisors, Dr Jake Kendrick and Dr Mubashar Hassan, for their help throughout this project."***



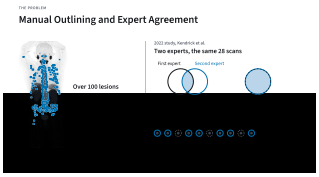
`Slide 2 · 80 s · ends at 1:40 · 47 words`

Metastatic Prostate Cancer and PSMA PET/CT

1. Purpose: Just enough medicine for a computing audience to follow the rest.
2. Numbers: 30% of patients relapse after treatment; five-year survival 35% or less once it has spread
3. Visual: Built: the survival figure; a cell with PSMA, then the tracer binding; PET, then CT beside it, then the two fused; lesion outlines; salivary glands, liver and kidney; a ring for the surrounding anatomy

Prostate cancer is common; when it relapses it can spread, and survival falls · PSMA is a protein on prostate cancer cells, and a radioactive tracer sticks to it · PET shows where the tracer gathers and CT shows the anatomy; lesions light up, but so do healthy organs

> `TRANSCRIPT TO READ · TIMED AT 95 S (TARGET 80 S)`
>
> ***"Prostate cancer is the second most common solid tumour in men worldwide, and once it spreads, the five-year survival rate drops to 35% or less. Finding the lesions is important, since the number of lesions and their location influence the treatment plan. PSMA stands for prostate-specific membrane antigen. It is a protein that is overexpressed on prostate cancer cells. A radioactive tracer that binds to PSMA is injected, and the PET scan shows where it collects. That is then paired with the CT scan, which shows the anatomy of the patient. PSMA PET/CT is the standard scan for finding where the disease has gone when it comes back. Lesions show up on the scan, but so do some healthy organs, such as the kidneys, salivary glands and liver, which take up the tracer normally. The challenge is to then train a model to find the lesions, but not get confused by normal uptake. That likely means it must be able to see the surrounding anatomy."***



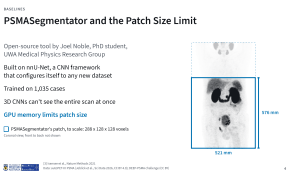
`Slide 3 · 40 s · ends at 2:20 · 38 words`

Manual Outlining and Expert Agreement

1. Purpose: State the problem early and simply.
2. Numbers: a second expert on 28 scans: outlines overlapped at Dice 33%; the second expert found 71.7% of the first expert's lesions
3. Visual: Built: a scan from this project with over 100 outlined lesions; the 2022 study apart on the right: the Dice pair, the identical pair, then ten lesion markers with seven found

Treatment depends on where the lesions are and how many there are · Outlining each one by hand is labour-intensive, worst in patients with the most disease · Experts disagree, so any automatic method is scored against an imperfect reference

> `TRANSCRIPT TO READ · TIMED AT 58 S (TARGET 40 S)`
>
> ***"Outlining the lesions in these scans manually is an extremely labour-intensive task. Some scans can have over 100 lesions, and the experts also disagree. In a 2022 study led by my supervisor, Dr Jake Kendrick, two experts independently outlined the same 28 scans. On average, their outlines overlapped with a Dice score of only 33%, where 100% would mean identical outlines. The second expert found about 7 out of 10 of the first expert's lesions. So automation reduces the manual work, but any automated method is still scored against these imperfect outlines. There's already a tool that does this, called PSMASegmentator."***



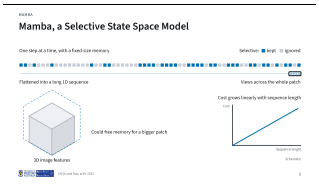
`Slide 4 · 60 s · ends at 3:20 · 51 words`

PSMASegmentator and the Patch Size Limit

1. Purpose: Show what already exists and the gap it leaves.
2. Numbers: PSMASegmentator trained on 1,035 scans, including every autoPET-III scan; the 2022 cascade found 73.0% of lesions on its own test set
3. Visual: Built: the name and lines in step with the transcript; PSMASegmentator's patch to scale with the rest dimmed, then it steps down the body

PSMASegmentator: an open-source tool from the UWA Medical Physics Research Group, built on nnU-Net, a self-configuring CNN · Before it, the group's 2022 two-stage nnU-Net cascade (Kendrick et al.), as context only · The gap: a 3D CNN reads the body one window at a time, and GPU memory caps the window size

> `TRANSCRIPT TO READ · TIMED AT 40 S (TARGET 60 S)`
>
> ***"PSMASegmentator is an open-source tool developed by Joel Noble, who is a PhD student at the UWA Medical Physics Research Group. It was built using the nnU-Net framework, which is a CNN framework that configures itself to any new dataset. It's been trained on 1,035 cases. However, one limit of 3D CNNs is that they can't see the entire scan at once, because GPU memory limits how big their patch, or window, can be. So that's where Mamba might be able to help."***



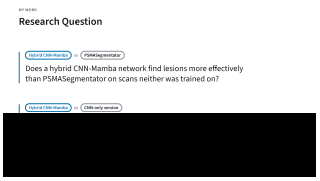
`Slide 5 · 50 s · ends at 4:10 · 55 words`

Mamba, a Selective State Space Model

1. Purpose: Mamba in one idea, why it might help, and why that needs testing.
2. Visual: Built: memory hops along a short sequence, keeping or ignoring each input; a 3D feature block flattens into a long sequence; the memory sweeps it; a linear cost line; a dashed bigger patch

Mamba (2023) reads a sequence once, in order, keeping a small memory and choosing what to keep · Its cost grows linearly with length; attention compares every pair, so its cost grows with the square · Hoped-for benefit: more of the scan per window for the same memory. Caution: a retest found U-Mamba's Mamba layers added nothing

> `TRANSCRIPT TO READ · TIMED AT 50 S (TARGET 50 S)`
>
> ***"Mamba reads a sequence one step at a time, carrying a fixed-size memory as it goes. It is selective, which means it chooses what to keep from each input and what to ignore. It was originally built for long sequences like text, which means for a 3D scan, the image features are flattened into a long 1D sequence. The idea is that it could help because it can view across the whole patch, while its cost grows linearly with sequence length. That could free memory for a bigger patch. That is exactly what I set out to test."***



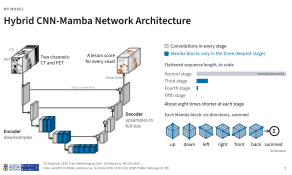
`Slide 6 · 20 s · ends at 4:30 · 31 words`

Research Question

1. Purpose: State exactly what was tested, as the introduction words it.
2. Visual: Built: the two questions in the transcript's words, each with its pair of models (the hybrid against PSMASegmentator; the hybrid against its CNN-only version)

Whether a hybrid CNN-Mamba network detects lesions better than PSMASegmentator on cases neither model trained on · Whether its Mamba blocks improve lesion detection over a CNN-only version of the same network

> `TRANSCRIPT TO READ · TIMED AT 25 S (TARGET 20 S)`
>
> ***"So my research question has two parts. Does a hybrid CNN-Mamba network find lesions more effectively than PSMASegmentator on scans that neither of them was trained on? And do the Mamba blocks really improve lesion detection, compared with a CNN-only version of the same network?"***



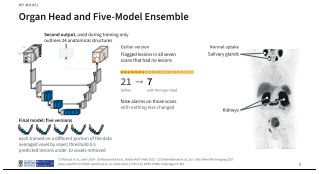
`Slide 7 · 70 s · ends at 5:40 · 47 words`

Hybrid CNN-Mamba Network Architecture

1. Purpose: Show the network and why Mamba sits where it does.
2. Numbers: five stages, 39.7M parameters; up to 27,648 tokens per Mamba block
3. Visual: Built: the dissertation's architecture figure assembled in layers (no patch or sequence sizes); flattened sequence length per stage, to scale; six reading directions, summed (schematic)

A U-shaped encoder-decoder: PET and CT in, a lesion mask out · Convolutions in the shallow stages, where detail is fine; Mamba blocks only in the deepest three, where sequences are short enough · Each Mamba block reads its tokens in six directions, forwards and backwards along each axis

> `TRANSCRIPT TO READ · TIMED AT 60 S (TARGET 70 S)`
>
> ***"So what I've decided to build is a U-shaped encoder-decoder with skip connections. The encoder has five stages, downsampling the scan between them, and the decoder upsamples the scan back to its full size. The input has two channels, the CT and the PET scan, and the output is a lesion score for every voxel, a voxel being a volumetric pixel. Convolutions run in every stage, but the Mamba blocks sit only in the three deepest stages, where the flattened sequences are shortest. Each step down makes the sequence about eight times shorter, so Mamba is cheap to run there. Each Mamba block reads the features in six directions: up, down, left, right, front and back. So every point gets information from every side."***



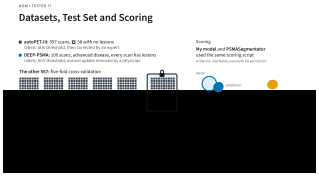
`Slide 8 · 60 s · ends at 6:40 · 45 words`

Organ Head and Five-Model Ensemble

1. Purpose: The choices that made the model work, each with its reason.
2. Numbers: 24 anatomical structures; false alarms on seven lesion-free scans: 21 before the organ head, 7 after; threshold 0.5; predicted lesions under 10 voxels removed
3. Visual: Built: the network carries over and its organ head comes forward; the scans with no lesions, all flagged, and their false alarms as a count; normal uptake on the scan of slide two; five versions of the network

Organ head: a second output learns anatomy during training only, so the network learns what normal uptake looks like · Final model: an ensemble of five versions, each trained on a different portion of the data, averaged voxel by voxel; predicted lesions under 10 voxels removed

> `TRANSCRIPT TO READ · TIMED AT 70 S (TARGET 60 S)`
>
> ***"An earlier version flagged lesions in all seven scans that had no lesions at all. The false alarms were typically in organs with normal physiological uptake, such as the kidneys and salivary glands. So, following PSMASegmentator's training recipe, which also learns organs, I added a second output to the network that learns to outline 24 anatomical structures. It's used during training only. With nothing else changed, false alarms on those seven scans fell from 21 to 7. For the final model, I trained an ensemble of five versions of this network, each on a different portion of the data, and averaged their predictions voxel by voxel. Any predicted lesion smaller than 10 voxels is then removed."***



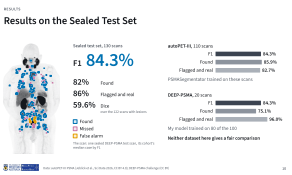
`Slide 9 · 60 s · ends at 7:40 · 69 words`

Datasets, Test Set and Scoring

1. Purpose: Give the numbers that follow their meaning.
2. Numbers: 697 scans: 597 autoPET-III and 100 DEEP-PSMA; 130 sealed test scans; 567 in five folds
3. Visual: Built: one tile per scan; the sealed test set put aside, the rest in five folds with their real mix; one scoring script; a lesion touched by a prediction ("found") beside a stray prediction ("false alarm"), schematic

Two public cohorts, autoPET-III and DEEP-PSMA; both start from an SUV threshold, then autoPET-III is corrected by an expert and DEEP-PSMA has normal uptake removed · A test set sealed away, never trained on and scored once; the rest split into folds for cross-validation · My model and PSMASegmentator scored by the same script; a lesion counts as found if the prediction touches it; F1 balances lesions found against false alarms

> `TRANSCRIPT TO READ · TIMED AT 70 S (TARGET 60 S)`
>
> ***"Two public datasets were used. The first is autoPET-III: 597 scans, 58 of which have no lesions at all. Its labels came from an SUV threshold, then were corrected by an expert. The second is DEEP-PSMA: 100 scans from patients with advanced disease, so every scan has lesions. Its labels also came from an SUV threshold, with normal uptake removed by a physician. For testing, 130 scans were sealed away, never trained on and only scored once. The rest went into five-fold cross-validation, which gave the five versions in the ensemble. For scoring, my model and PSMASegmentator used the same script. A lesion counts as found if any predicted voxel touches it. F1 balances lesions found against false alarms, and Dice shows how well the outlines match."***



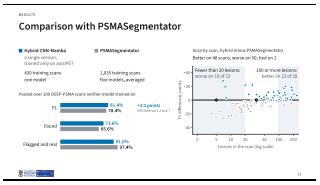
`Slide 10 · 60 s · ends at 8:40 · 38 words`

Results on the Sealed Test Set

1. Purpose: The headline result, on scans held back to the end.
2. Numbers: F1 84.3%, precision 86.2%, sensitivity 82.4%, Dice 59.6% over 130 scans; autoPET-III sensitivity 85.9%; DEEP-PSMA precision 96.0%
3. Visual: Built: the qualitative figure's DEEP-PSMA test scan with found, missed and false alarm in its three colours; the headline numbers; the two cohorts' bars (cohort numbers on screen only)

Scored once, on scans no model had trained on · Same F1 on both cohorts, but a different trade: autoPET-III found more, DEEP-PSMA was more precise · It trained on both cohorts, so it is not compared with PSMASegmentator here

> `TRANSCRIPT TO READ · TIMED AT 67 S (TARGET 60 S)`
>
> ***"So, how did the final ensemble model do? On the sealed test set, the ensemble scored an F1 of 84.3%. It found 82% of the lesions, and 86% of what it flagged were real lesions, with a Dice of 59.6%. The F1 was the same on both datasets, but the balance was different. On autoPET, it found more of the lesions, but raised more false alarms. On DEEP-PSMA, it was the other way round: almost everything it flagged was real, but it missed more lesions. And PSMASegmentator was actually trained on the autoPET test scans, and my model was trained on most of DEEP-PSMA, so neither dataset here could give a fair comparison. For that, I needed a dataset that neither model had been trained on."***



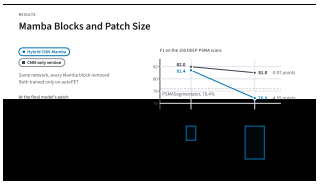
`Slide 11 · 60 s · ends at 9:40 · 55 words`

Comparison with PSMASegmentator

1. Purpose: The fair head-to-head: the first research question.
2. Numbers: 100 scans; pooled F1 81.4% against 78.4%, +3.0 points (1.2 to 4.7); sensitivity +8.1 points, precision -6.4 points; per-scan F1 -0.6 points (-2.8 to 1.6)
3. Visual: Built: the two models with the setup as said; pooled bars (F1, found, flagged and real) for the hybrid and PSMASegmentator; then one dot per scan, hybrid minus PSMASegmentator against lesions in the scan, with the bands for few and many lesions

A single version of the model, trained only on autoPET-III, against PSMASegmentator on the DEEP-PSMA scans neither model trained on · Higher pooled F1, despite most of the setup favouring PSMASegmentator; it found more lesions, at the cost of more false alarms · Per scan, mixed: better on scans with many lesions, worse on scans with few

> `TRANSCRIPT TO READ · TIMED AT 65 S (TARGET 60 S)`
>
> ***"So I took a single version of my model, trained only on autoPET, and scored it against PSMASegmentator on all 100 DEEP-PSMA scans, which neither model had seen. My model scored an F1 of 81.4%, against 78.4% for PSMASegmentator, so about three points higher. And that's despite most of the setup favouring PSMASegmentator: it was trained on more than twice as many scans, and averages five models, against my one. My model found more of the lesions, but raised more false alarms. Scan by scan, it was mixed: better on scans with many lesions, worse on scans with few. But pooled over all the scans, my model scored higher. So, is that down to Mamba?"***



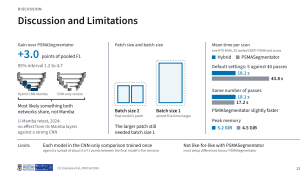
`Slide 12 · 50 s · ends at 10:30 · 73 words`

Mamba Blocks and Patch Size

1. Purpose: The second research question: is Mamba the reason?
2. Numbers: smaller patch, hybrid minus CNN-only: -0.58 on DEEP-PSMA (no clear effect), +1.79 on autoPET-III; CNN-only F1 82.0 against PSMASegmentator 78.4%; larger patch: hybrid -4.55, CNN-only -0.97 F1 points; difference -3.57 (-4.72 to -2.48)
3. Visual: Built: the hybrid chip, and the CNN-only version sliding out from behind it; F1 on DEEP-PSMA at the final model's patch and at the larger one, each patch drawn to scale under its column; PSMASegmentator as a dashed line; the hybrid drops, the CNN-only version barely moves

The hybrid and its CNN-only version (every Mamba block removed), both trained only on autoPET-III, each at the final model's patch and at a patch almost five times larger · At the smaller patch the Mamba blocks added little; the CNN-only version also scored above PSMASegmentator, so the gain is not credited to Mamba · On DEEP-PSMA the larger patch hurt the hybrid far more than the CNN-only version, the opposite of what was expected

> `TRANSCRIPT TO READ · TIMED AT 75 S (TARGET 50 S)`
>
> ***"To answer the second question, I trained the same network without its Mamba blocks, a CNN-only version, also trained only on autoPET. I trained both at the final model's patch size, and at a patch almost five times larger. At the smaller patch, the Mamba blocks added little: there was no clear difference on DEEP-PSMA, and a gain of under two points on autoPET. And the CNN-only version also scored above PSMASegmentator on DEEP-PSMA, 82.0% against 78.4%, so the gain over PSMASegmentator can't be credited to Mamba specifically. The larger patch was where Mamba was meant to help. Instead, on the unseen DEEP-PSMA scans, it cost the hybrid about four and a half F1 points, and the CNN-only version only about one: the opposite of what I'd expected. So what does this really mean?"***



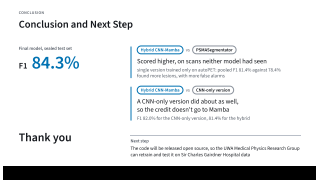
`Slide 13 · 60 s · ends at 11:30 · 71 words`

Discussion and Limitations

1. Purpose: What the results mean, and their limits.
2. Numbers: default settings: 18.1 s against 43.8 s per scan; same passes: 18.1 s against 17.2 s; peak memory 5.2 GiB against 4.5 GiB
3. Visual: Built: the pooled gain with its interval; the hybrid and the CNN-only version as network pictures, joined; the U-Mamba retest; the patches to scale in one training batch each; time per scan, where PSMASegmentator's bar shrinks at the same number of passes; the limits along the bottom

The gain over PSMASegmentator most likely comes from something both networks share, not Mamba; a 2024 retest of U-Mamba found the same · Mamba's linear cost did not pay off: the larger patch still forced batch size 1 for both networks, and at the same number of passes PSMASegmentator was slightly faster and lighter · Limits: one training run per model in the CNN-only comparison, a smaller CNN-only version, not like-for-like with PSMASegmentator

> `TRANSCRIPT TO READ`
>
> ***"It means the pooled gain over PSMASegmentator is real, but it most likely comes from something both of my networks share, not from Mamba. A 2024 retest of U-Mamba, another Mamba network, found the same: its Mamba layers had no effect once it was compared with a strong CNN. Mamba's linear cost also didn't pay off in practice. The larger patch still forced a batch size of one. My abstract says my model takes about half PSMASegmentator's time per scan, but that's only because PSMASegmentator runs test-time augmentation, with eight times as many passes. Here, with the same number of passes, PSMASegmentator was slightly faster on average, and used slightly less memory. The main limits are that each model in the CNN-only comparison was trained only once, and the comparison with PSMASegmentator isn't like-for-like."***



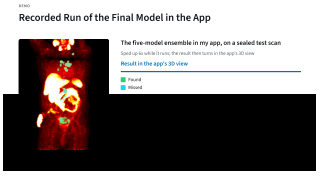
`Slide 14 · 30 s · ends at 12:00 · 55 words`

Conclusion and Next Step

1. Purpose: One message to leave with, the next step, then questions.
2. Visual: Built: the test-set F1; the research question's two pairs of models from slide six, each with its answer from the transcript; the next step; thank you

The ensemble reached F1 84.3% on a sealed test set · On unseen scans, the autoPET-only hybrid beat PSMASegmentator in pooled F1, and the CNN-only version did about as well · Next step: the code released open source, so the UWA Medical Physics Research Group can retrain and score it on the Sir Charles Gairdner Hospital cohort

> `TRANSCRIPT TO READ`
>
> ***"To wrap up: on the sealed test set, my final model scored an F1 of 84.3%. On scans neither model had seen, my network scored higher than PSMASegmentator overall, but a CNN-only version did about as well, so the credit doesn't go to Mamba. Next, the code will be released open source, so the UWA Medical Physics Research Group can retrain and test it on Sir Charles Gairdner Hospital data. Thank you, and I'm happy to take any questions."***



`Backup B1 · 0 words`

Recorded Run of the Final Model in the App

1. Leave up during questions: the real recorded run, looping

No transcript.

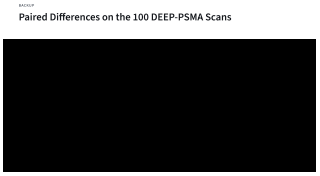


`Backup B2 · 0 words`

Setup Differences with PSMASegmentator

1. Question: Is the comparison with PSMASegmentator like-for-like?

No transcript.

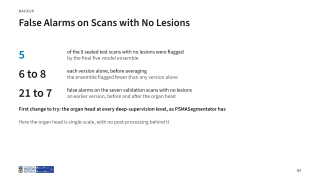


`Backup B3 · 0 words`

Paired Differences on the 100 DEEP-PSMA Scans

1. Question: Is the gain in pooled F1 real, and what happens scan by scan?

No transcript.

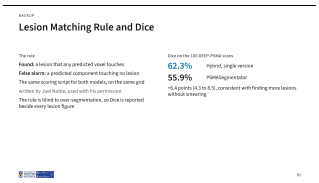


Backup B4 · 0 words

False Alarms on Scans with No Lesions

1. Question: Does it raise false alarms on scans with no lesions?

No transcript.

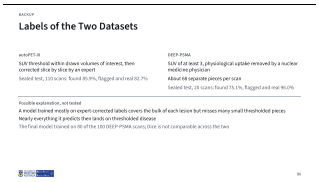


Backup B5 · 0 words

Lesion Matching Rule and Dice

1. Question: Why count a lesion as found if any predicted voxel touches it?

No transcript.

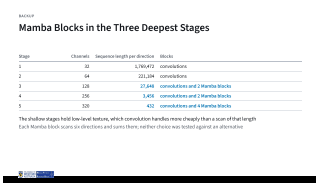


Backup B6 · 0 words

Labels of the Two Datasets

1. Question: Why is precision so high on DEEP-PSMA, and how were the datasets labelled?

No transcript.

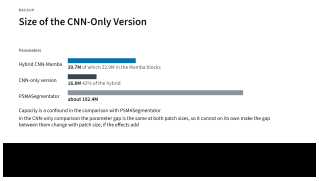


Backup B7 · 0 words

Mamba Blocks in the Three Deepest Stages

1. Question: Why put Mamba only in the three deepest stages?

No transcript.

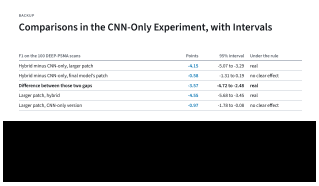


Backup B8 · 0 words

Size of the CNN-Only Version

1. Question: Is the CNN-only version just a smaller network?

No transcript.



Backup B9 · 0 words

Comparisons in the CNN-Only Experiment, with Intervals

1. Question: Each model was trained once, the larger patch at batch size 1. Is the result just noise?

No transcript.



Backup B10 · 0 words

Inference Time in the Dissertation and the Abstract

1. Question: The abstract says about half the time. Why do the numbers differ?

No transcript.

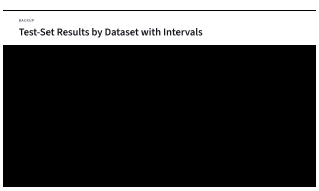


Backup B11 · 0 words

Cross-Validation and the Five Folds

1. Question: How was cross-validation done?

No transcript.



Backup B12 · 0 words

Test-Set Results by Dataset with Intervals

1. Question: What are the confidence intervals on the test results?

No transcript.


BACKUP

References

[1] J. Kleesiek, K. J. Kurz, G. M. Himper, P. Weerkamp-Bartsch, J. S. L. Ding, and M. A. Ebert, "Deep MRI brain extraction: A 3D convolutional neural network for skull stripping," arXiv preprint arXiv:1603.04413, 2016.

[2] S. Shakeri-Zadeh, A. Attar-Khorasani, M. Eslimi et al., "Prostate and pitfalls in clinical interpretation of prostate-specific membrane antigen (PSMA)-targeted PET imaging," *Journal of Nuclear Medicine and Radiation Therapy*, vol. 8, no. 1, pp. 1–10, 2017.

[3] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation," *Nature Methods*, vol. 18, no. 2, pp. 203–211, 2021.

[4] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," arXiv preprint arXiv:2312.00752, 2023.

[5] J. Ding, Y. Xu, S. Tang, G. Cai, R. Xie, S. Wu, F. Gao, and L. Zhu, "Hyphformer: A Long-range segmentation modeling Hybrid(to-global) 3-D medical image segmentor," *IEEE Transactions on Medical Imaging*, vol. 41, no. 1, pp. 6–15, 2021.

[6] J. Ding, Y. Xu, S. Tang, G. Liu, and L. Zhu, "Hyphformer: Long-range sequential modeling Hybrid(to-global) 3-D medical image segmentor," in *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2021, 2021.

[7] N. Antonio, B. Hiroshi, Y. Kirchhoff, S. Vitor, C. Ulrich, K. H. Maier-Hein, and F. Isensee, "Automated segmentation of head and neck tumors: Multimodal fusion methods for tumor segmentation in PET/CT imaging," in *HN-segment at MICCAI/HECKTOR*, 2024.

[8] J. Nawrocki, H. C. Shen, M. T. Bryan, M. Pisetska, D. Hinds, A. M. Eaton, T. Heye, D. T. Roth, J. Cyriac, S. Yang, N. Rai, Jun R. Sripriya, "Histopreparatory: Histo-preparation of LN anatomic structures in CT images," *Radiology: AIHI Scientific Reports*, vol. 3, no. 1, p. 1, 2024.

[9] A. Ahlrich, B. Taha Emre, A. Mittermeier et al., "nnU-Net-Only PSMA-CT/T1 Dataset with manually annotated tumor lesions," *Scientific Data*, vol. 11, p. 1321, 2024.

[10] P. Jackson, J. M. Hinds, J. Bonaus, and N. Kirchner, "Deep learning evaluation for enhanced pregnancy – prostate specific membrane antigen (DEEP-PSMA)," *Challenge Proceedings (Biomedical Image Analysis Challenges Initiative)*, Oracle, 2024, not peer-reviewed.

[11] F. Isensee, T. Wald, C. Ulrich, M. Baumgartner, S. Roy, K. Maier-Hein, and P. F. Jaeger, "nnU-Net revisited: A call for rigorous validation in 3D medical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI 2024)*, 2024.

Backup B13 · 27 words

References

These are the works cited on the slides, in the order they appear, in the same IEEE style as the dissertation. The dissertation has the full list.